HSLU Hochschule Luzern

How to Structure a Scientific Report / Paper

Prof. Dr. Umberto Michelucci

Computer Science Department
Applied AI Center

Reproducibility and Replicability (two opposite definitions)

"**Reproducibility**" refers to instances in which the original researcher's data and computer codes are used to regenerate the results, while "**replicability**" refers to instances in which a researcher collects new data to arrive at the same scientific findings as a previous study.

"**Replicability**" refers to independent researchers arriving at the same results using their own data and methods, while "**reproducibility**" refers to a different team arriving at the same results using the original author's artifacts.

National Academies of Sciences, Engineering, and Medicine. "Reproducibility and replicability in science." (2019).

Open Science and **Reproducibility** Promoting Open Science helps researchers

1. to more easily and reliably build on previous work through Open Access publications and openly available data
2. to accelerate discovery through less repetition of inaccessible work and of data already produced for other projects
3. to increase and maintain the trustworthiness of scientific results which can be verified and replicated

https://openscience-summerschool.ch/assets/downloads/lectures/osss-2022-presentation-furrer.pdf

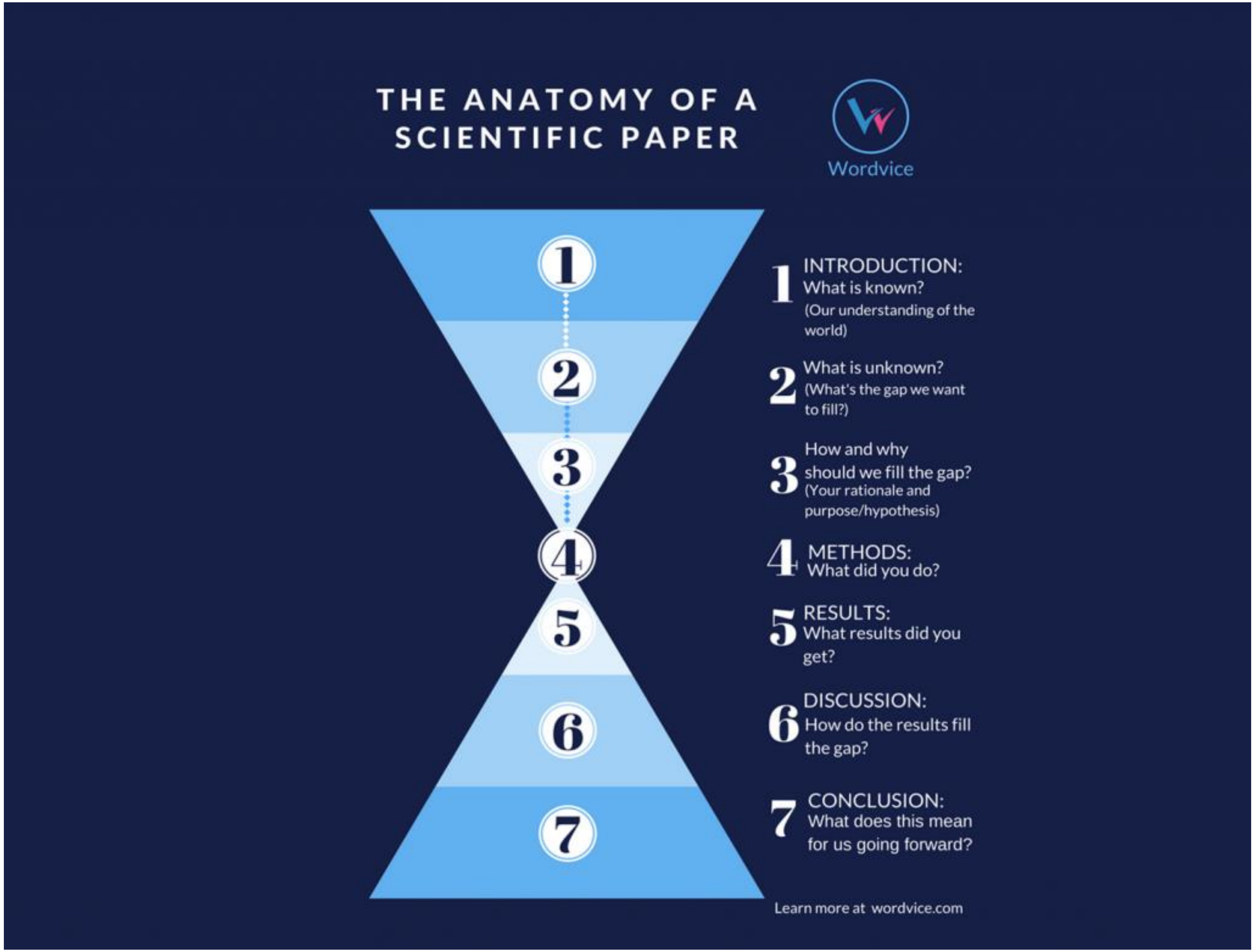
Your task as scientists

One important task, as scientists, is to make sure that your work is <u>reproducible</u>. If is not, your work is scientifically useless.

In addition your work must be <u>transparent.</u> All the steps must be clear and understandable. Another researcher must be able to re-implement (or understand) everything you did (every single step) from scratch.

Anatomy of a Scientific Paper

1. Introduction
2. Methods
3. Results
4. Discussion
5. Conclusion



https://blog.wordvice.com/seminar-how-to-write-an-effective-research-paper/

Introduction

it should contain

- the problem statement,
- literature review (state ofthe art),
- contributions (can be in separate sections).

It should answer the questions ***what?, why is this relevant?, what did other do already? and finally why should the reader care?***

Methods

it should contain

- the description of what you did,
- how you did it,
- What data you used

NOT THE RESULTS.

TIP: *If you realise that explaining all the details will make your paper too long, do not omit information. Simply publish it in an Appendix or in additional material related to your paper.*

Results and Discussion

here you should list

- all your results as numbers, tables, plots, etc.

TIP: Discussion on implications should not be here,

Conclusions

Here you should discuss

- the implications of your results,
- Their limitations
- possible further research possibilities.

TIP: You should be very honest especially about limitations.

HSLU Hochschule Luzern

Literature Review

a.k.a. State of the art

4. Mai 2026

FH Zentralschweiz

Literature Review: Goal

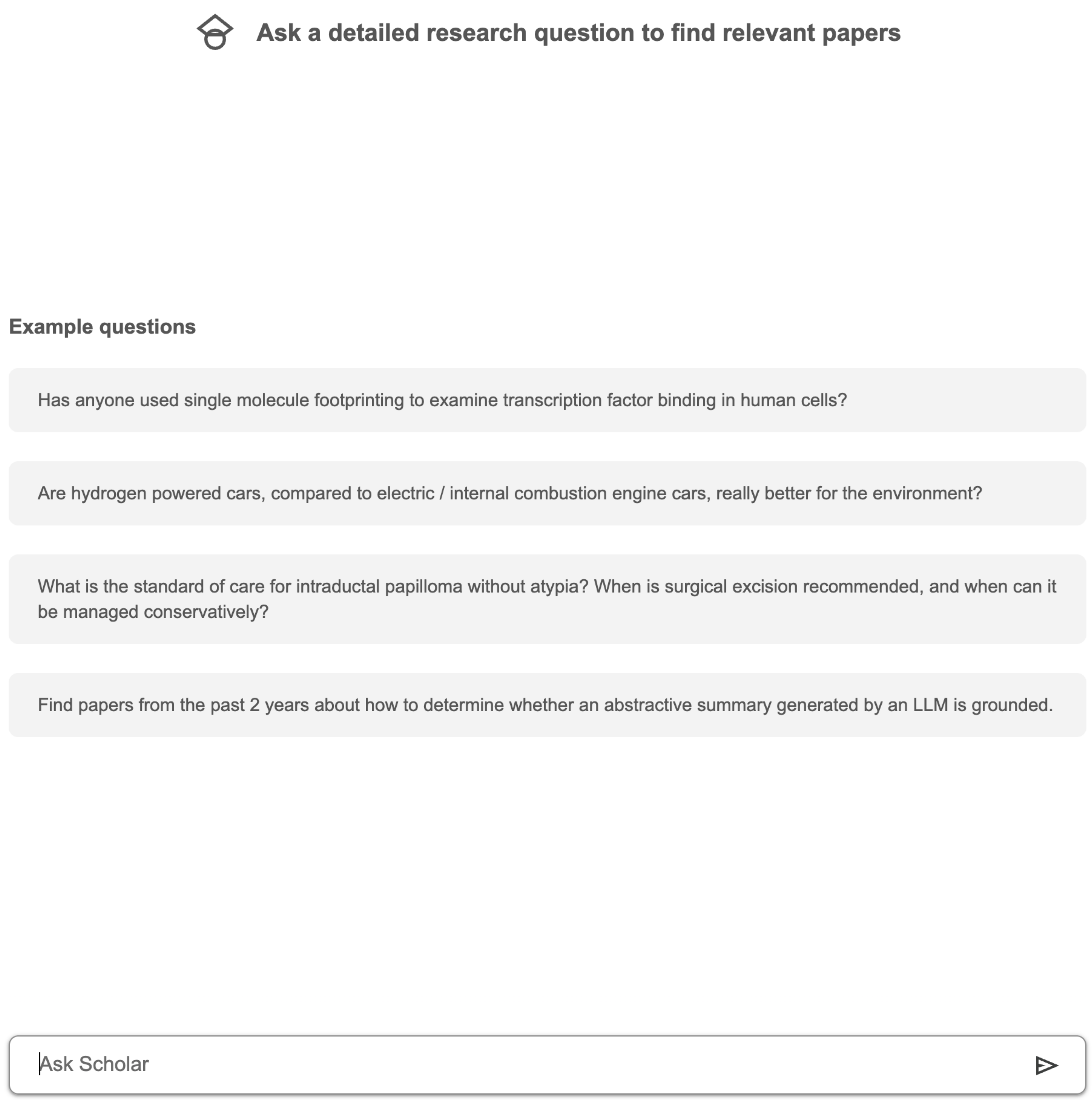
The main goal os literature review is

to evalute the state of the common knowledge on your problem (and your research questions).

This helps you in assessing what is already known and decide if your idea is novel or has been done already

Literature Review: Tools

https://scholar.google.com/ (and especially "labs")



Literature Review: Additional Tools

https://arxiv.org/

https://pubmed.ncbi.nlm.nih.gov/ (more for medicine)

Tools to organise references: Zotero https://www.zotero.org/ (tools for mac and windows exist).

HSLU Hochschule Luzern

Sources and References

Reasons of having references

Compilation of knowledge already gained / of what knowledge is available in relation to the research question.

Prove that you know the state of research.

To show where one's own question comes from.

Prove your own statements

Adopt successful concepts, definitions and formulation

Through a careful outline of the state of the art in research, the scientific relevance of one's own research approach is demonstrated.

Citation Styles

Style	In-Text	Reference Section Entry
ACM	[1]	[1] J. Smith. 2023. *The Future of AI*. Tech Press, New York.
IEEE	[1]	[1] J. Smith, *The Future of AI*. New York, NY: Tech Press, 2023.
APA	(Smith, 2023)	Smith, J. (2023). *The future of AI*. Tech Press.
Chicago	(Smith 2023)	Smith, John. 2023. *The Future of AI*. New York: Tech Press.
Harvard	(Smith, 2023)	Smith, J. (2023) *The Future of AI*. New York: Tech Press.

https://owl.purdue.edu/owl/research_and_citation/apa6_style/apa_formatting_and_style_guide/in_text_citations_the_basics.html
https://libguides.murdoch.edu.au/IEEE/all

Citation Styles - Example

However, a systematic review of the field reveals critical limitations in current SSC methodologies when applied to scientific novelty detection. The literature is predominantly driven by constraint-based methods utilizing must-link and cannot-link constraints (Basu et al., 2004b; Jiang et al., 2013). While effective for enhancing cluster purity, these methods typically operate under the assumption that the provided supervision is "representative" of the underlying data distribution (González-Almagro et al., 2023). This assumption forces

References

Tiba Zaki Abdulhameed, Suhad A Yousif, Venus W Samawi, and Hasnaa Imad Al-Shaikhli. Ss-dbscan: Semi-supervised density-based spatial clustering of applications with noise for meaningful clustering in diverse density data. *IEEE Access*, 12:131507–131520, 2024.

Mihael Ankerst, Markus M Breunig, Hans-Peter Kriegel, and Jörg Sander. Optics: Ordering points to identify the clustering structure. *ACM Sigmod record*, 28(2):49–60, 1999.

M Baghshah and S Shouraki. Kernel-based metric learning for semi-supervised clustering. *Neurocomputing*, 2010. Based on Source 212.

https://owl.purdue.edu/owl/research_and_citation/apa6_style/apa_formatting_and_style_guide/in_text_citations_the_basics.html
https://libguides.murdoch.edu.au/IEEE/all

Online Gratis Kurs

https://www.hslu.ch/de-ch/informatik/studium/wissenschaftliches-arbeiten-am-departement-informatik/

Wissenschaftliches Arbeiten am Departement Informatik Analytisch und anwendungsorientiert recherchieren und dokumentieren